

HSTW PROJECT –3

TESLA STOCK ANALYSIS REPORT

1. INTRODUCTION

- **Core Business:** Tesla, Inc. is recognized as a global leader in electric vehicles (EV) and sustainable energy solutions.
- **Market Dynamics:** The stock is characterized by significant price fluctuations influenced by technological innovation, production milestones, and global expansion.
- **Analysis Importance:** Stock analysis allows investors to decode growth patterns, risk levels, and the underlying market sentiment toward disruptive industries.
- **Macro Factors:** Performance is heavily tied to macroeconomic factors and internal corporate announcements.

2. OBJECTIVES OF THE ANALYSIS

- **Trend Tracking:** Monitoring the historical trajectory of Tesla's closing prices over a 12-year span.
- **Growth Identification:** Pinpointing specific periods marked by rapid bullish growth and sharp bearish declines.
- **Liquidity & Risk:** Analyzing trading volume to assess market liquidity and evaluating volatility to measure investment risk.
- **Sentiment Mapping:** Quantifying investor behavior through the ratio of bullish to bearish trading days.

3. Price Trends & 30-Day Moving Average

- **Theory of Moving Averages:** The 30-day Moving Average (MA_30) smooths out "noise" (short-term volatility) to highlight the primary trend.

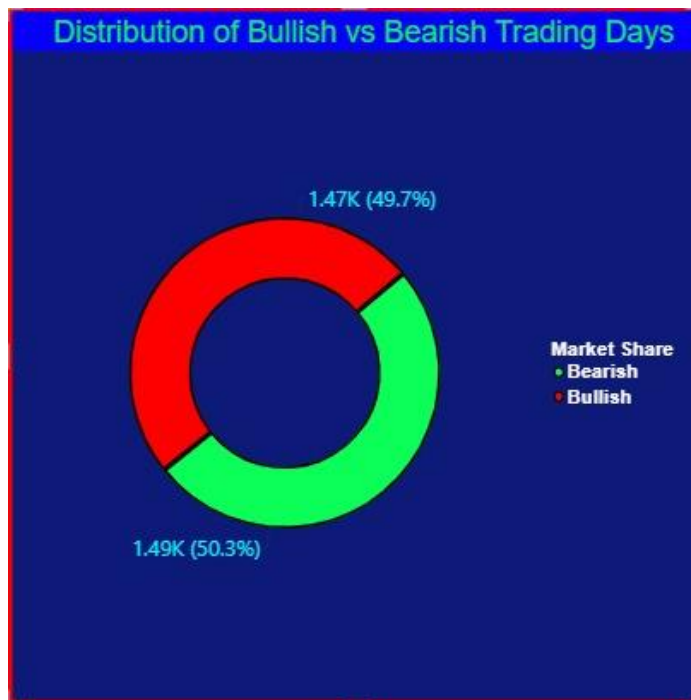
- **Growth Indicators:** An upward-sloping MA_30 line indicates sustained long-term growth and high investor confidence.
- **Market Corrections:** Sudden drops in the price line below the moving average often signal market corrections or external shocks.
- **Visual Analysis:** The chart shows a massive breakout starting in late 2020, reaching peaks above \$1,200.



Tesla Closing Price Trend (2019–2022) highlighting the 30-Day Moving Average as a trend-following indicator.

4. Distribution of Bullish vs. Bearish Days

- **Metric Definition:** A "Bullish Day" is recorded when the Closing Price exceeds the Opening Price.
- **Statistical Breakdown:** Out of the analyzed period, 1.49K days (50.3%) were bullish, while 1.47K days (49.7%) were bearish.
- **Sentiment Insight:** The near 50/50 split suggests that while the stock has grown immensely, it experiences frequent daily "tug-of-wars" between buyers and sellers.
- **Market Share:** This visual represents the "Market Share" of daily price direction.



A pie chart illustrating the balanced distribution of positive (bullish) and negative (bearish) trading days.

5. Annual Trading Volume & Liquidity Analysis

- **Volume as a Signal:** High trading volume confirms the strength behind a price move.
- **Bullish Momentum:** Rising prices accompanied by high volume indicate strong buying interest.
- **Panic Selling:** High volume during a price drop indicates mass exits or "panic selling".
- **Observation:** The year 2020 saw an unprecedented spike in trading volume, coinciding with Tesla's rapid price appreciation.



Annual Trading Volume Trend (2010–2022) identifying peaks in market liquidity and investor participation.

6. Volatility Indicator

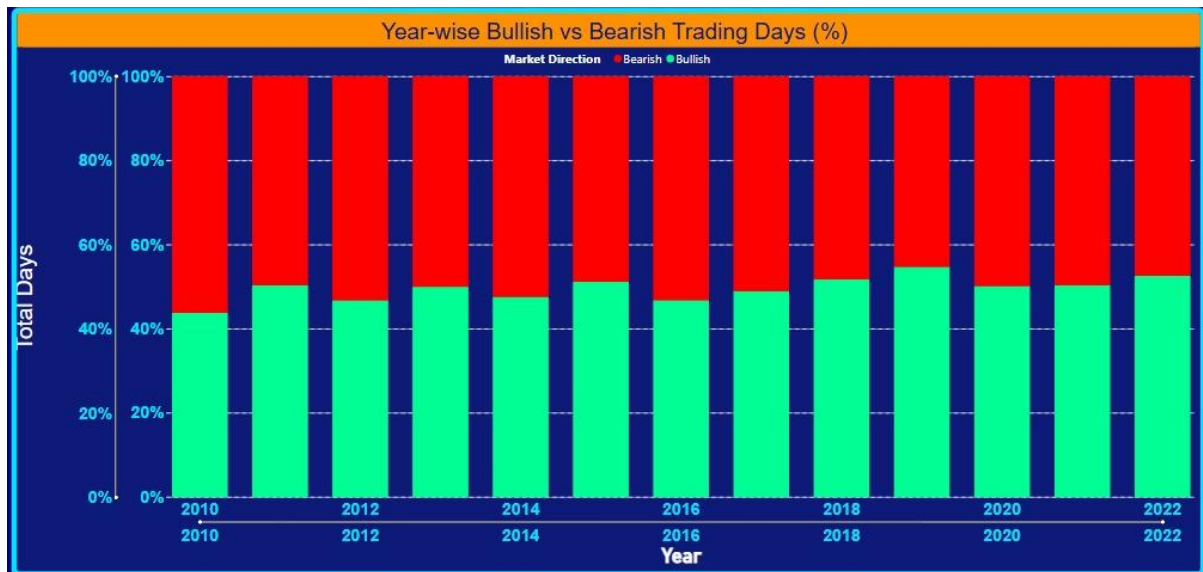
- **Calculating Volatility:** The Daily Price Range is defined as the difference between the High Price and Low Price for a single day.
- **Risk Measurement:** This visual is the primary tool for measuring the "inner-day" risk of the stock.
- **Trend in Volatility:** The analysis shows that as Tesla's price increased (2020–2022), the daily price range expanded significantly, indicating higher risk for short-term traders.



Average Daily Price Range by Year, showcasing the escalation of stock volatility during Tesla's high-growth phases.

7. Year-wise Bullish vs. Bearish Trends (%)

- **Annual Sentiment:** This 100% stacked bar chart compares the ratio of "up" days to "down" days for each year.
- **Consistency:** Despite price growth, the ratio of bullish to bearish days remains relatively stable across most years .
- **Market Direction:** This helps identify specific years where the market direction was more decisively bullish (e.g., 2020–2021) versus correction years.



Year-wise Bullish vs. Bearish Trading Days percentage (2010–2022) to visualize annual sentiment shifts.

8. Integrated Dashboard

- **Key Performance Indicators (KPIs):** The dashboard summarizes Max Close, Min Close, Average Close, and Total Volume.
- **Interactivity:** Using filters for Day, Month, Quarter, and Year allows for granular inspection of specific events.
- **Data Consolidation:** Combining price trends, volume, and volatility in one view provides a holistic understanding of market health.



Comprehensive Tesla Stock Analysis Dashboard for multi-dimensional data visualization.

9. CONCLUSION

- **Growth Profile:** Tesla is a classic example of a high-growth, high-volatility asset.
- **Speculative Behavior:** Significant correction periods reflect the stock's sensitivity to market news and speculative trading.
- **Investor Insight:** While long-term potential is high, the stock's volatile nature requires rigorous risk management strategies.
- **Final Word:** Trading volume remains a key indicator of investor participation during major market events